6. Transformers

Two circuits; one changing magnetic field

Commissioning a two-winding magnetic system

Lesson 6 · two sessions · learner: isolated battery source, 12 V DC maximum

BOUNDARY

Use only the pictured two-cell pulse circuit. No outlet, wall adapter, opened appliance, inverter, ignition coil, charged capacitor, or large battery pack. An adult checks every unpowered connection and controls stop decisions.

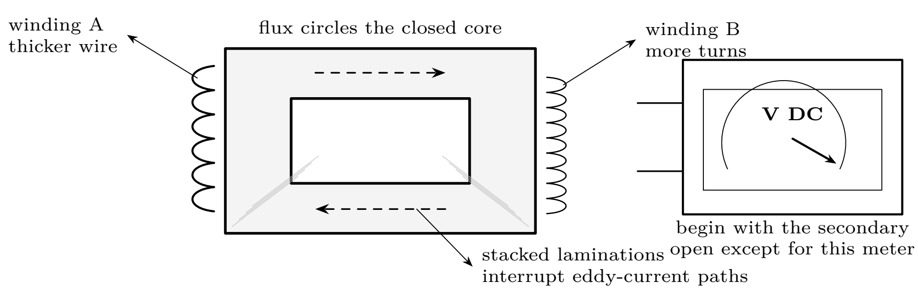
What you need

- Closed laminated core; two intact insulated low-voltage windings
- 2 AA cells in holder; momentary normally-open pushbutton
- Two digital multimeters; insulated clip leads; tape labels
- One 220 Ω, 0.25 W resistor for the final load test

Predict first

A secondary has three times as many turns as the primary. On a quick press, should its pulse be smaller, equal, or larger? While the button remains held, what should the secondary meter settle toward?

Survey the magnetic circuit



COLD INSPECTION

Battery holder out. Inspect the winding insulation and core edges. Identify each pair of winding leads by continuity, measure its resistance, and verify open circuit from every lead to the metal core. A core-contact reading ends the investigation.

Winding	lead labels	turns	resistance	core isolated?	first inference
A				yes / stop	
B				yes / stop	

Inspection inference. Resistance can help distinguish the windings, but it does not directly reveal turns because wire gauge and length both matter. Which observation supports your label choice? ______________________

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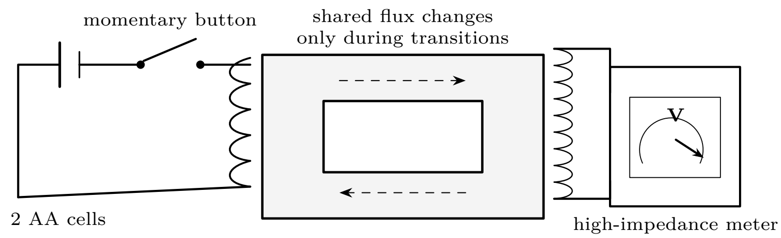
Commissioning a two-winding magnetic system

Lesson 6A · make, hold, break · change one connection only while unpowered

Wire the pulse test



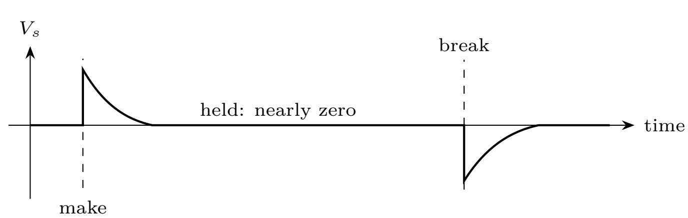
1. Set the secondary meter to DC volts and its highest suitable range.
2. Tap the button and note the first deflection: this is **make**. Observe the reading while held very briefly, then release for **break**.
3. Repeat three times. Record signs as well as magnitudes.
4. Reverse only the secondary leads and repeat. Predict before tapping.
5. With the battery removed, open the core joint slightly; reconnect and repeat one cycle. Restore the closed core afterward.

Configuration	make peak	held value	break peak	repeat spread	conclusion
closed core					
secondary reversed					
core gap					

Interrogate the evidence

Why does release reverse the pulse? Which observation proves that a transformer responds to changing flux rather than merely to primary current? Did reversing the meter leads alter energy transfer, or only the displayed polarity? Why did a narrow core gap matter?

Sketch the time record



6. Transformers

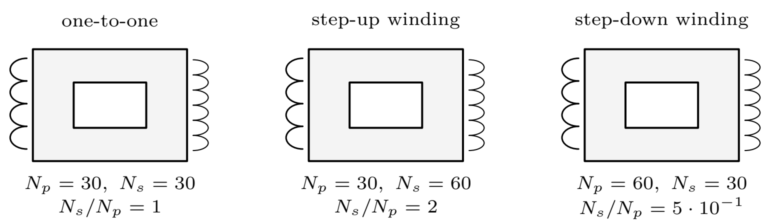
Turns and loading

Commissioning a two-winding magnetic system

Lesson 6B · measure ratios · use consistent, brief switch pulses

Map turns to voltage

Choose one winding as the primary. If taps or interchangeable coils are available, compare at least three secondary turn counts without changing the core or primary.



$$\text{pulse-voltage ratio} \quad \frac{|V_s|}{|V_p|} \approx \frac{N_s}{N_p}$$

N_p	N_s	turns ratio	primary peak	predicted secondary	measured secondary
30	30	1.00			
30	60	2.00			
60	30	0.50			

Worked forecast

With $N_p = 40$, $N_s = 120$, and a measured primary transition of 2.2 V, the ideal prediction is

$$|V_s| = 2.2\,\mathrm{V}\left(\frac{120}{40}\right) = 6.6\,\mathrm{V}.$$

If the secondary pulse supplies 8 mA, ideal power balance gives $I_p \approx I_s(N_s/N_p) = 24\,\mathrm{mA}$. Voltage can rise only as available current falls.

Ratio drill before measuring

Complete the ideal prediction, then circle whether the winding is being used as a step-up, one-to-one, or step-down pair.

N_p	N_s	V_p	ideal V_s	ideal I_p/I_s	classification
50	150	1.6 V			up / same / down
120	60	2.4 V			up / same / down
80	80	2.0 V			up / same / down

Model check. Doubling N_s should double pulse magnitude.

Support: ________________ Challenge: ________________

6. Transformers

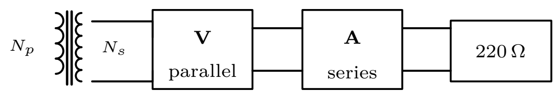
Load characterization

Commissioning a two-winding magnetic system

Lesson 6B.2 · voltage and current · meter-only test before known resistor

Apply a controlled load

load branch: connect only after open-circuit test



1. Predict $I_s = V_s/R_L$, then have the adult verify meter jack positions.
2. Measure one open-circuit pulse. Remove the battery, add the resistor and series ammeter, then repeat the same brief switching motion.
3. Record loaded V_s, I_s, and V_sI_s. Disconnect immediately.

Test	V_s	I_s	output product	change from open circuit	evidence of loss
open		—	—	baseline	
loaded					

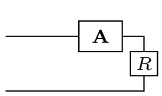
Reconcile voltage, current, and power

For the loaded trial, calculate

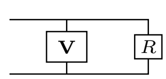
$$I_{\text{predicted}} = \frac{V_{\text{open}}}{R_L}, \qquad P_s = V_{\text{loaded}} I_s, \qquad \text{regulation drop} = \frac{V_{\text{open}} - V_{\text{loaded}}}{V_{\text{open}}} \times 100\%.$$

Quantity	prediction	measurement	difference	what could explain it?
secondary current				
secondary power				
voltage drop				

Meter-placement audit



correct: A in series



correct: V in parallel

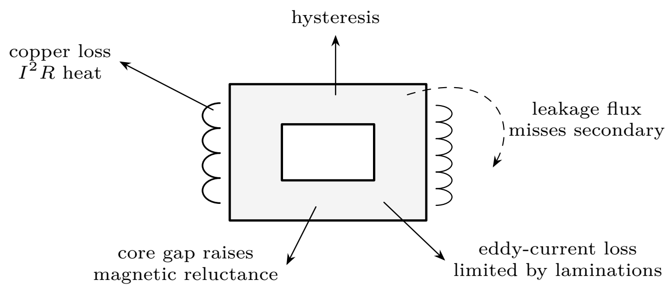
An ammeter placed across the winding is nearly a short circuit. Remove the battery before moving either lead. The adult verifies both meter jacks before every energized reading.

Predict first

If the resistor value doubles, predict the direction of change for secondary current, terminal-voltage sag, and winding heating. Which quantity should stay closest to its open-circuit value?

Lesson 6C · explain discrepancies · all diagnosis begins unpowered

Account for what the ideal ratio omits



Copper Long, fine wire has resistance. Load current produces heat and terminal-voltage sag.

Core Magnetization lags the field; induced currents circulate within each lamination. Both consume input energy.

Coupling Some primary flux does not link every secondary turn. A loose joint or physical gap makes this worse.

Measurement Hand switching is not a calibrated waveform. Compare trends and polarity; do not claim precision from short-lived meter peaks.

Fault tree

Symptom	Unpowered check first	Distinguish these causes
No secondary pulse	continuity, meter mode/range, lead seating	open winding; wrong terminals; failed switch
Weak on every ratio	close and clamp core; inspect joint; repeat motion	core gap; loose coupling; inconsistent switching
Correct open, poor loaded	verify resistor; confirm ammeter is series	excess load; winding resistance; meter burden
Reading changes when flexed	inspect and gently test each terminal	broken strand; incomplete enamel removal
Heat, odor, discoloration	remove battery; stop the lesson	shorted turn; held button; wiring error
Continuity to core	do not energize	damaged insulation

ISOLATION ORDER

Remove cells. Look and smell. Verify winding continuity. Verify isolation to core. Check meter leads, jacks, and range. Restore the meter-only test. Add the known resistor last. Never diagnose heat by applying power again.

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Evidence hearing

Commissioning a two-winding magnetic system

Lesson 6C.2 · defend each claim · use measurements, not definitions

Final evidence hearing

Defend each statement with a measurement, not a definition.

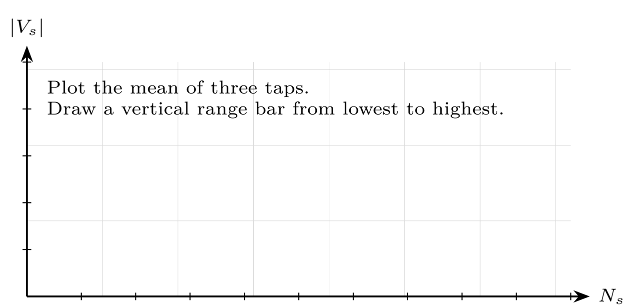
Statement	Exhibit	Reasoning
Flux must change		
Polarity follows change		
Turns govern voltage		
Load changes input demand		
Real devices dissipate		

Exit questions

Why can a 3:1 voltage increase not also give a 3:1 current increase? Which single comparison isolates the effect of the core? What meter mistake would place nearly a short circuit across a winding? Which loss best explains warm wire, and which two belong chiefly to the core?

Controlled extension

Plot $|V_s|$ against N_s with primary, core, range, and switch motion fixed. Add error bars from three repeated taps. State whether the graph supports proportionality and name the largest uncertainty.



Graph decision	Evidence in the pattern	Largest uncertainty
proportional / not proportional		

FINAL RELEASE

Before storing the apparatus: remove the cells, detach the load, return both meters to their voltage jacks, and inspect the windings for warmth or damage. Adult initials: ________ Date: ________